

Assignment no. 1 (v.1.0¹): IPv6 security

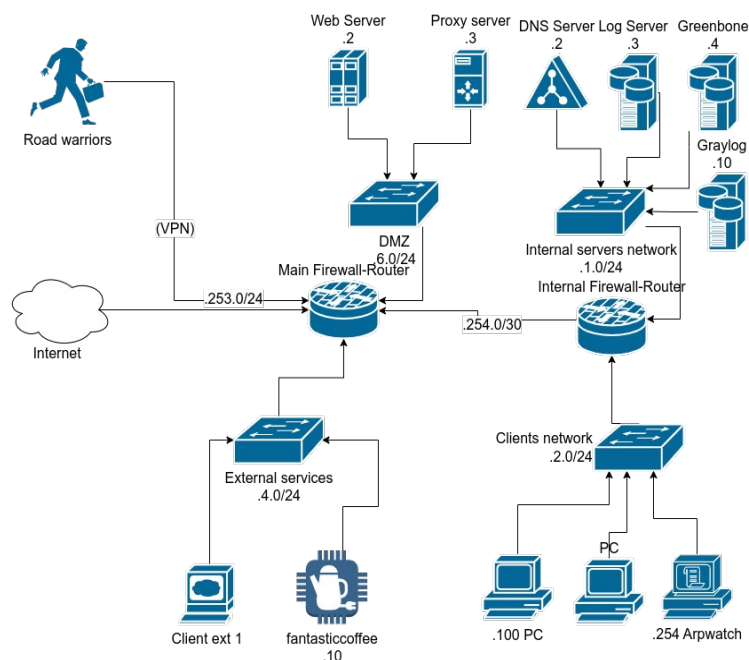
In this assignment, your team must properly configure the ACME network to use IPv6 in the virtual environment assigned to the team and complete the exercises to experiment with IPv6 security.

While performing your assignment, your team also has to describe the process you have followed and the type of tests you have performed to check your assignment and verify the performed security tests.

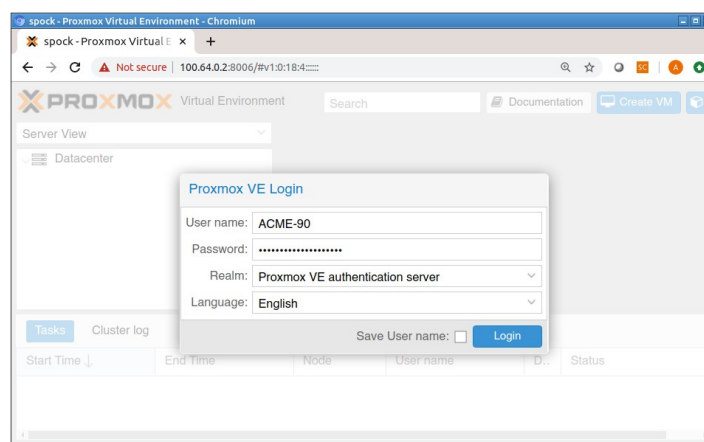
If you suspect something is wrong or not as expected, please comment on the Classroom page so that all the students can see and possibly agree or disagree.

The ACME network

The ACME network consists of several (virtual) host machines, running within the Proxmox environment, namely a platform for enterprise virtualization. The topology of the network is the following:

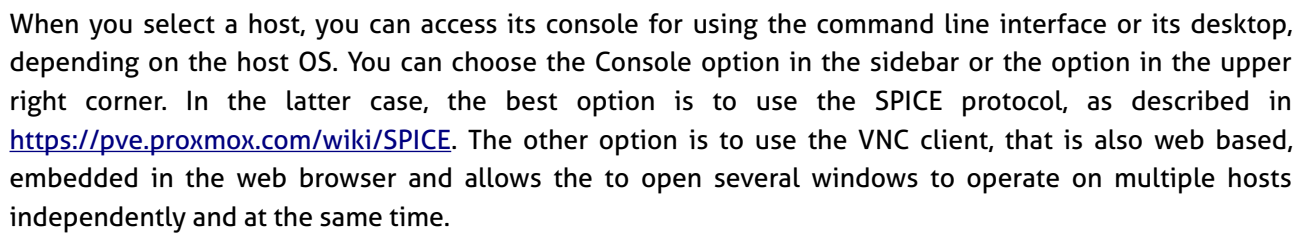


Your ACME network is accessible through the VPN you have received by email. When starting the VPN you will be granted the access to the <https://100.64.0.2:8006/> URI, where you can access your Proxmox panel, as in the following figure:



When you enter your credentials, you will be able to access all the hosts in the networks. The control panel of Proxmox is represented in the following figure:

1) Last change: 27/03/26



IPv6 configuration

The procedure shown in the video is just a possible approach: if you think there are better arrangements, you are welcome to explore and follow them.

The second task of this assignment is to experiment with the IPv6 security within the ACME network. Your team must replicate the [course exercises](#) provided by the [IPv6 Security Training Course of RIPE NCC](#). In particular, you have to replicate the exercises for scanning the network and those targeting the NDP (Neighbor Discovery Protocol). You can ignore the exercises for MLD and traffic filtering.

Your team must detail the process to perform the exercises and collect evidence about their completion (e.g., collection of pcap files, screenshots and alike), which must be included in the final report.

You have to prepare a document that reports the activities you have performed to realize the assignment tasks. The document should be named `ACME_XX_a1_report.pdf` and should be included together with any additional material in **one single** .zip file named `ACME_XX_a1.zip`, without any subdirectory, if not needed (i.e., testing scripts). The zip file is the **only thing** your group has to submit in **classroom**, and only **one** of the members.

1. Group number
2. Student names and numbers

3. Initial brainstorming (where you write your considerations about what to do and how)
4. Evaluation of the IPv6 address assignment tasks and its finalization
5. IPv6 security exercises implementation
6. Final remarks

Hand-in dates

Remember: you must submit the assignment before taking the written exam. The evaluation will consider completeness, clearness, and proficiency.